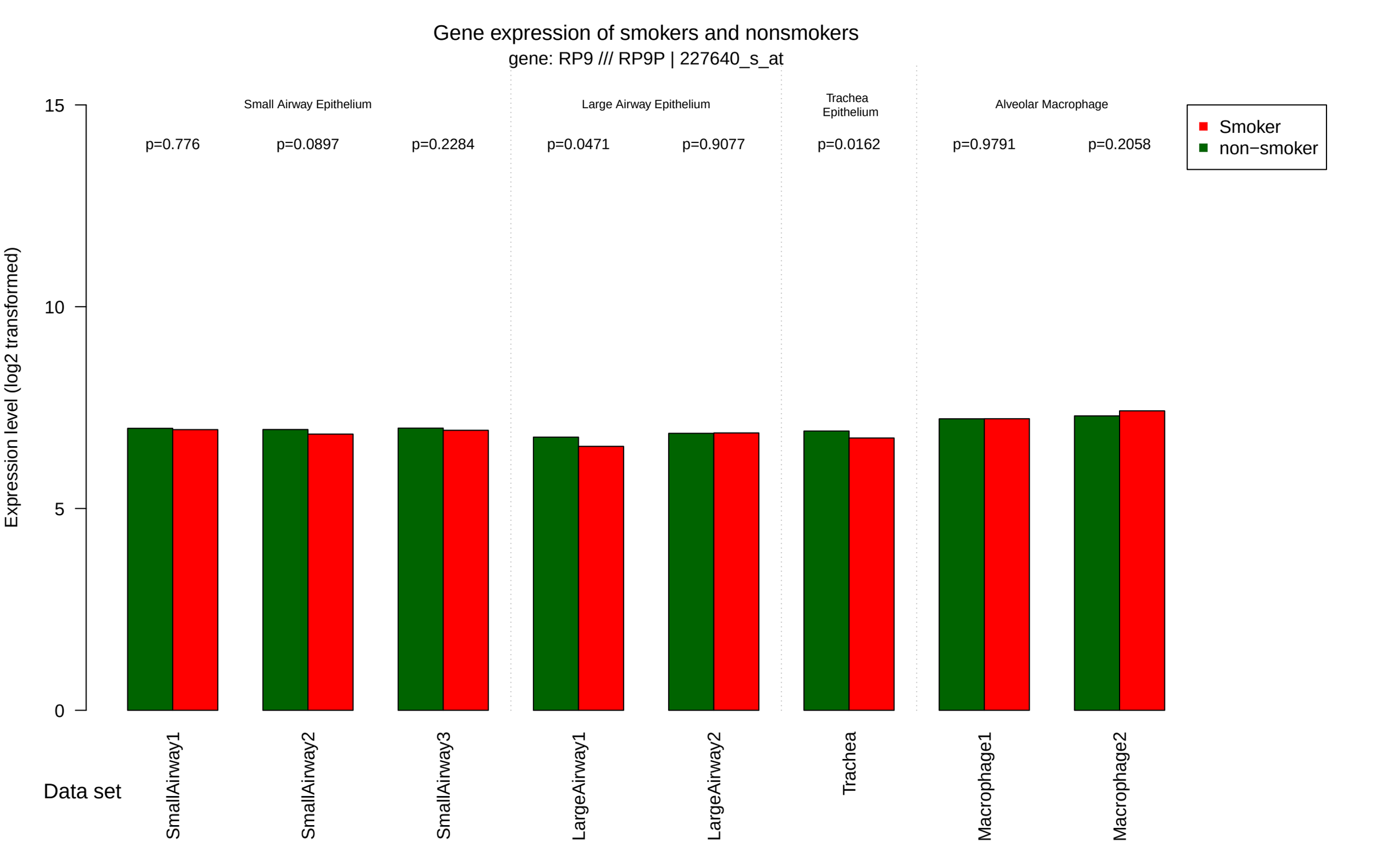
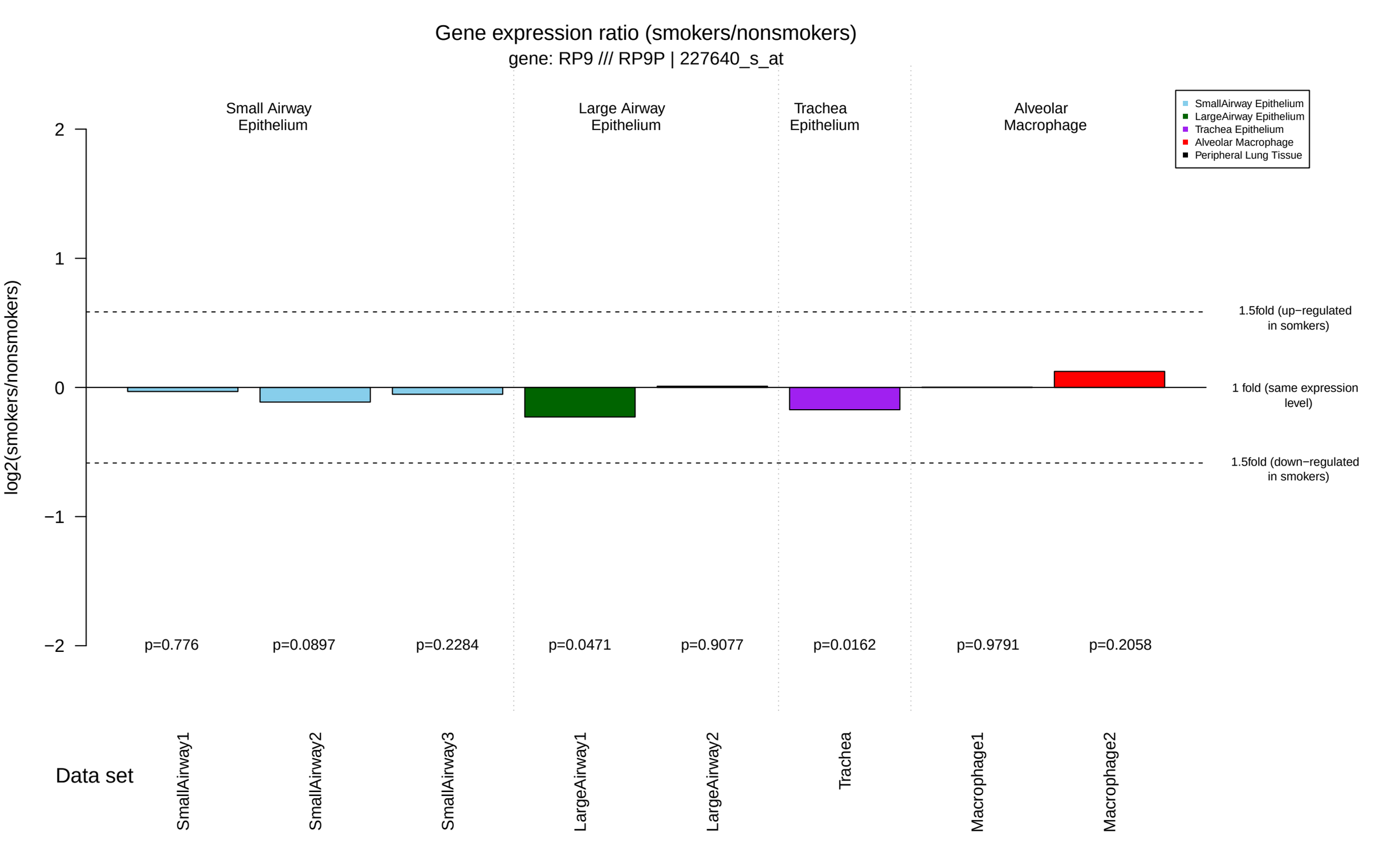
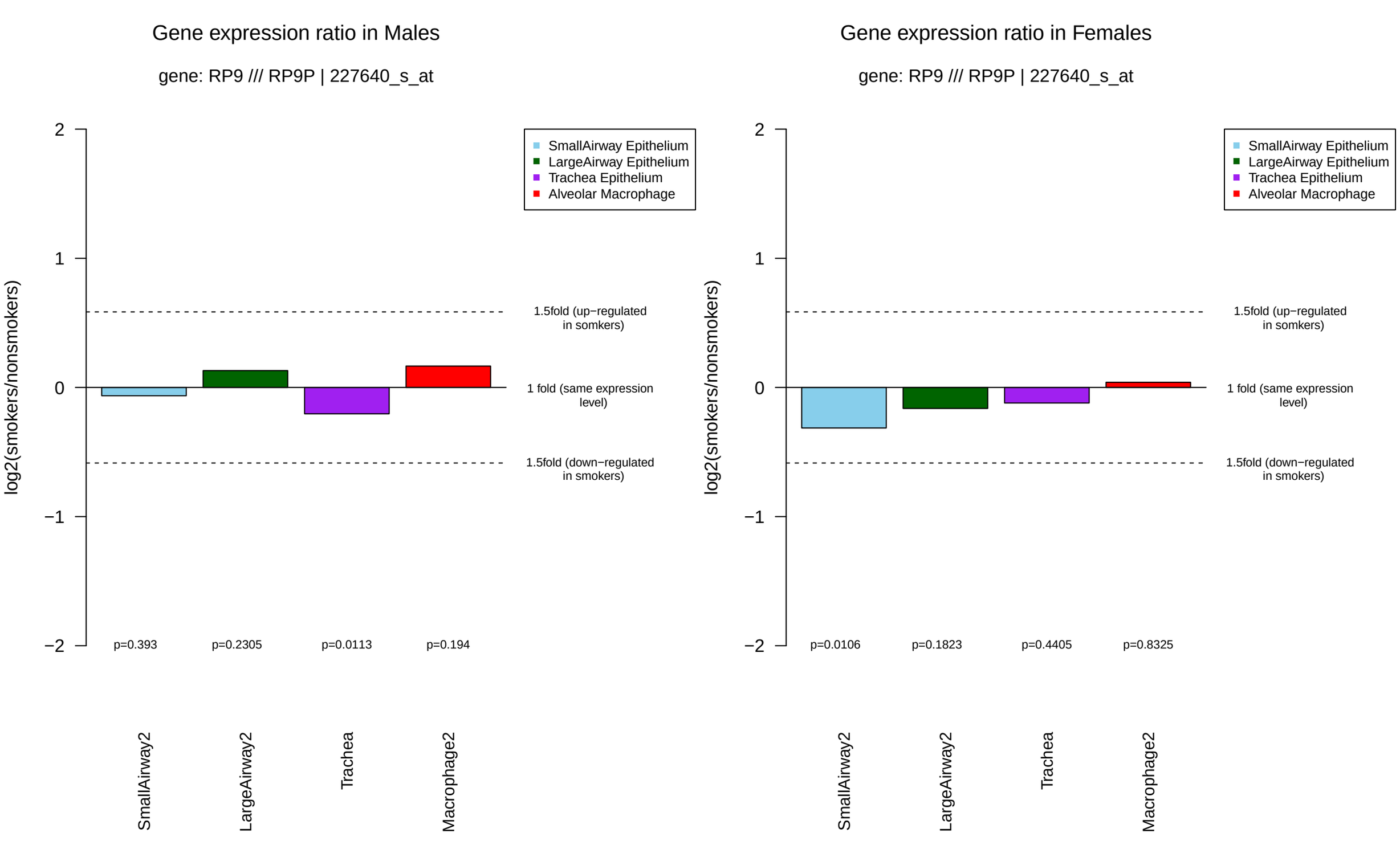
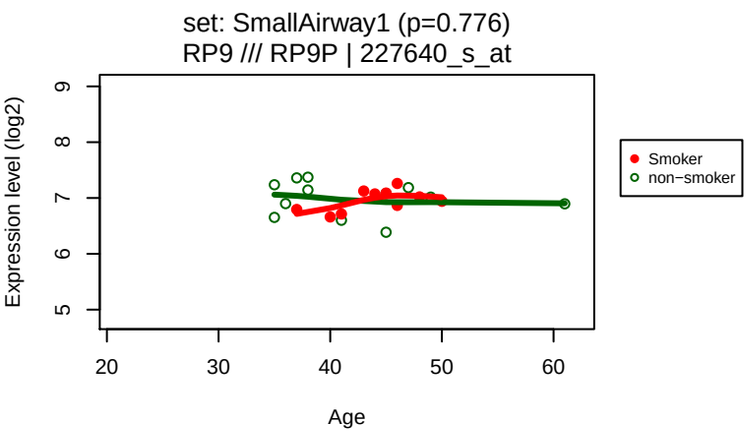
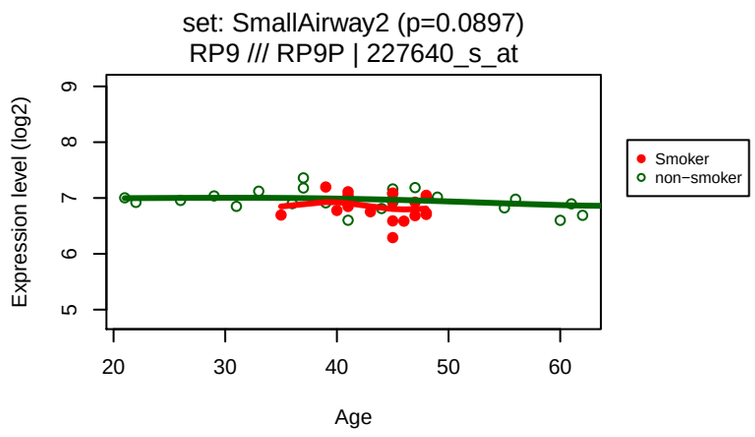
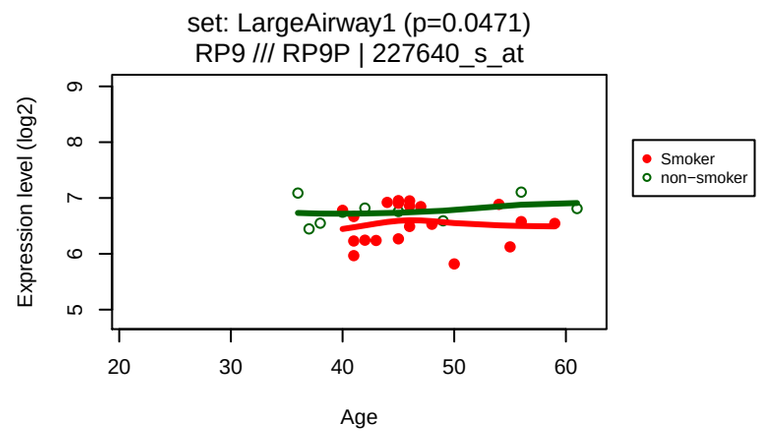

Gene expression of smokers and nonsmokers
gene: RP9 /// RP9P | 227640_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.776
p=0.0897
p=0.2284
p=0.0471
p=0.9077
p=0.0162
p=0.9791
p=0.2058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

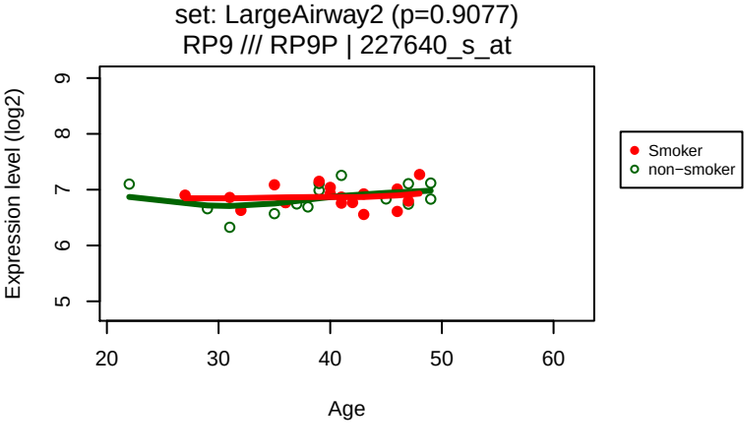

Gene expression ratio (smokers/nonsmokers)
gene: RP9 /// RP9P | 227640_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.776
p=0.0897
p=0.2284
p=0.0471
p=0.9077
p=0.0162
p=0.9791
p=0.2058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

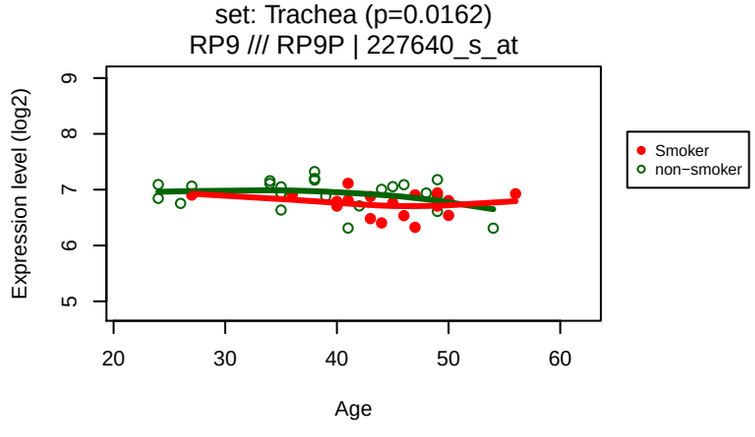

Gene expression ratio in Males
gene: RP9 /// RP9P | 227640_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.393
p=0.2305
p=0.0113
p=0.194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RP9 /// RP9P | 227640_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0106
p=0.1823
p=0.4405
p=0.8325
SmallAirway2
LargeAirway2
Trachea
Macrophage2

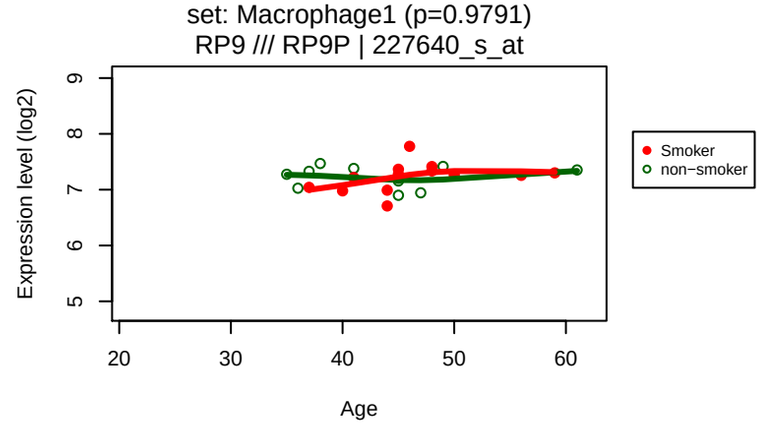

set: SmallAirway1 (p=0.776)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker

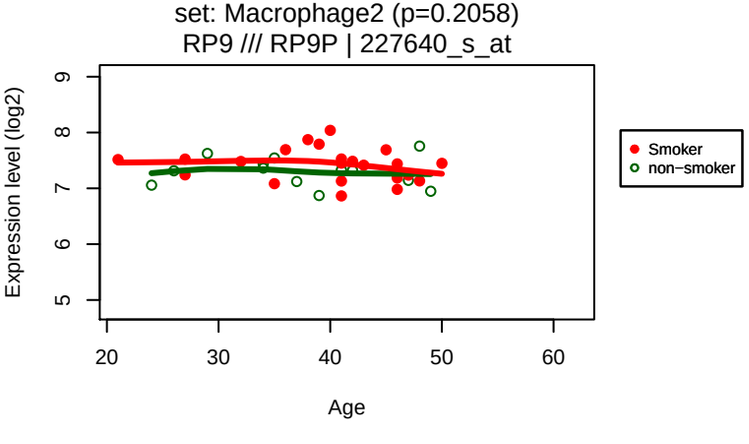

set: SmallAirway2 (p=0.0897)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker

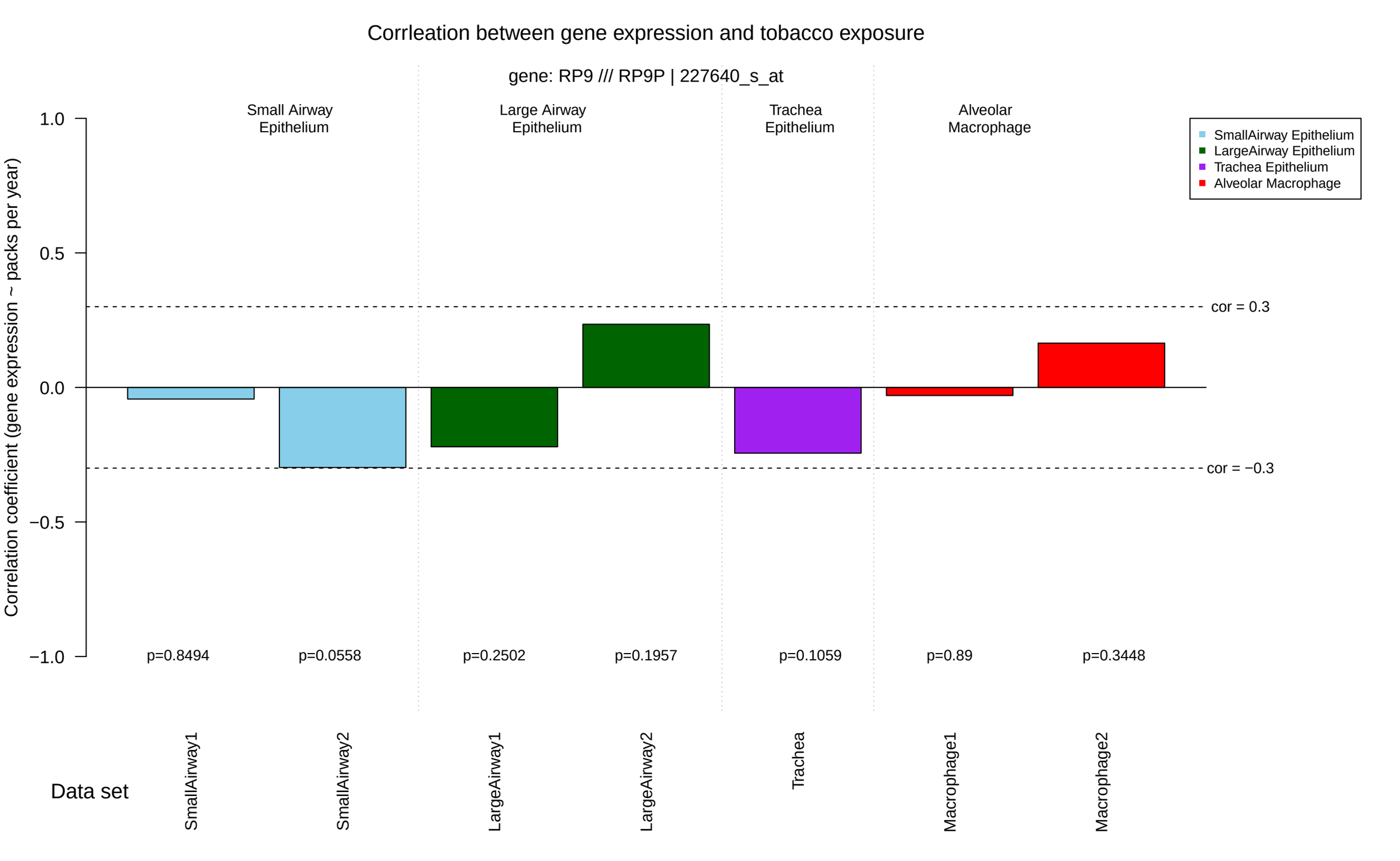

set: LargeAirway1 (p=0.0471)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9077)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0162)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9791)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2058)
RP9 /// RP9P | 227640_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RP9 /// RP9P | 227640_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8494
p=0.0558
p=0.2502
p=0.1957
p=0.1059
p=0.89
p=0.3448
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2